

1.0 Fundamentals of computing networks

Learning Points

- ☐ Understand the fundamentals of data communication and computer networks
 - ☐ Understand the OSI model and the functions of its seven layers
 - ☐ Understand how data is transmitted at the Physical Layer
 - ☐ Understand how the Data Link Layer provides node-to-node communication
 - ☐ Understand the functions of the Network, Transport, Session, Presentation and Application layers
 - ☐ Understand data encapsulation and decapsulation across the OSI layers
-

Data communications

The exchange of data between two devices via some form of transmission

characteristics of effective communications systems:

- Delivery - correct destination
- accuracy - data integrity
- Timeliness - on time
- Jitter - packet arrival variation

/ * protocol - a set of rules which govern data communications between two devices (TBC in notes 2 or 3)

Network

two or more devices connected together for the purpose of communication

/ * a standalone is a non connected device

/ ** device can consist of hosts (servers and laptops) AND connecting devices (routers and switches)

Advantages of networks

- file sharing
- hardware sharing
- communications between devices
- Roaming access (access of files from any devices with correct authorization)
- centralized maintenance and updating
- user monitoring

Disadvantages of networks

- cost of setup
- additional maintenance costs
- spread of malware
- hacking - unauthorized access of information

Types of networks

- Local area networks - geographically constrained to a single site
- Wide area networks - spans over wide geographical area / multiple sites
- internetwork / internet - two or more connected LAN / WAN
- Internet - global system of interconnected computer networks using the TCP/IP
- Intranet - private network within an organization, only for use by members

OSI model - Open system interconnection(developed by the International Organisation for Standardization)

no.	Layer	Function	Data Type
7	Application	interfaces directly with use end applications	data
6	Presentation	provides services for the application layer - Translates data between systems, encrypts, compression, encoding	data
5	Session	Establish maintain and terminate sessions between applications	data
4	Transport	End to end communication error recovery and flow control	segments
3	Network	Manages routing of data packets between devices across different networks	packets
2	Data Link	Ensures reliable data transfer across physical link, framing and error detection	frames
1	Physical	Transmission of bitstream over a medium	bits

Each layer provides services to the upper levels, while allowing each layer to function independently from the last

Physical Layer

coordination of procedures and functions required to carry a bit stream over a physical medium

- Decides type of Interface to use (mechanical or electrical)
- Procedures and functions physical devices have to carry out

Full Functionality

- represents all data as bits to output as signal
- control data rate
- ensure synchronized bit-rate
- type of line configurations used
- physical topology
- Transmission mode (half vs full duplex)

Physical mediums

- twisted copper wire - to reduce interference of signals
- coaxial cable - transmit high frequency (single core)
- fibre optic - transfer via reflection of light in fibreglass

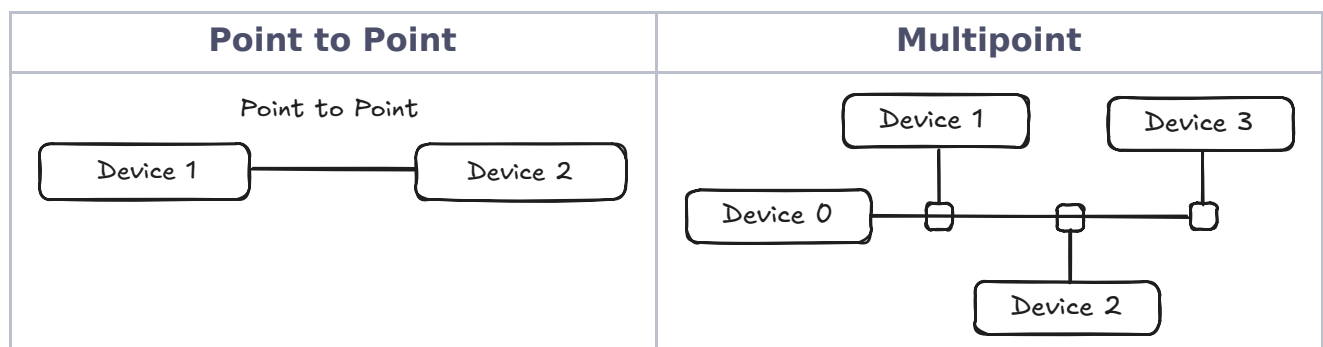
Bits are represented in analog through continuously varying voltage, radio waves...ect

Line configurations

point to point - a direct connection between two devices

multi-point / multi-drop - more than one device shares the same link

- space share - simultaneously use the same link
- time share - take turns using link



Physical topologies

How devices are connected to form network(geometrically)

- mesh - each device has dedicated p2p with other devices
- star - each device has dedicated p2p with central controller

- bus - multi-point where devices are connected by a long wire
- ring - each device has a p2p with the device on either side, data transfer is one directional

Transmission mode - direction of transfer between two devices

- simplex - unidirectional, one transmits and one receives
- half duplex - both can transfer and receive but not at same time (time share)
- full duplex - both cant transfer and receive at the same time (space share)

Data Link Layer

ensures direct communication between two devices on the same local network

- oversees packet delivery between two devices

Full functionality

- physical addressing / framing - encapsulation of data into frames



- trailer - error detection - crc, checksum, parity bits ...ect.
- header - physical address / MAC address of next device or final destination
- flow control
- error detection
- access control - decides what device has control over a link if two or more devices are in a connection
 - Media Access Control (MAC) - Listen in before transmitting
 - Carrier Sense Multiple access with collision avoidance(CSMA/CA): wait for signal before transmission

Data Link Layers

Logical Link control (LLC)

- Interface with upper layer, error check, flow control

Media Access Control (MAC)

- channel access via MAC and CSMA
- physical addressing

MAC address - written in hexadecimal of 6 groups of 2 digits

- 48 bytes total
- separators denoted by '.' OR '-'



OUI

Device specific ID

- OUI (organizationally unique identifier) - manufacturer ID

* MAC address is hard-coded to the Network interface card

Network Layer

Oversees source to destination delivery across multiple networks.

* network layer does not concern the relation between packets

** layer is only relevant if cross boundary communication is needed

- logical addressing: prepend header of packet with logical address of sender and receiver
- routing: allows creation of inter-networks through cross network addressing

Transport Layer

Controls process to process delivery of message.

- service point addressing - delivery to correct programs through port addressing
- Segmentation and reassembly - separates message to be sent, reassembles packets received
- Connection control
 - establishes connection and terminates connection for each communication
 - treats packets as independent parcels
- Flow control - end to end flow control of entire connection
- Error control - error control for entire message (end to end)

Session Layer

Allows presentation to organize communication for multiple sessions.

- session initiation and tear-down - create and terminate session on transfer start and end
- token management: half duplex - which entity owns token and authorization to transmit
- Session to transport connection mapping: what data goes to what session and vice versa

Presentation Layer

Responsible for presentation of data to the application layer.

- provides various services to application layer:
 - encryption
 - compression
 - translation ect.

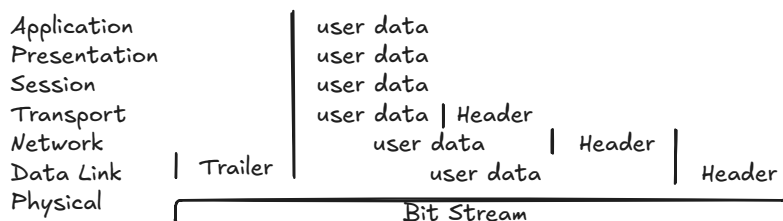
Application Layer:

definition of user end services:

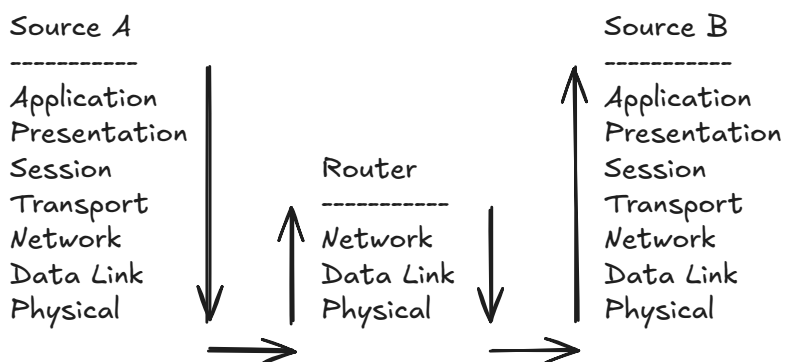
- identification of services
- Qos parameters for application
- security mechanisms
- synchronization of communication applications

Data Encapsulation and Decapsulation

encapsulation



data transport



During encapsulation, data from the top layer is treated as a single data block

Each layer encapsulates data with the necessary information:

- transport adds sequence numbers and acknowledgements (TCP)
- network source and next hop logical addresses

- data link next hop physical address and trailer